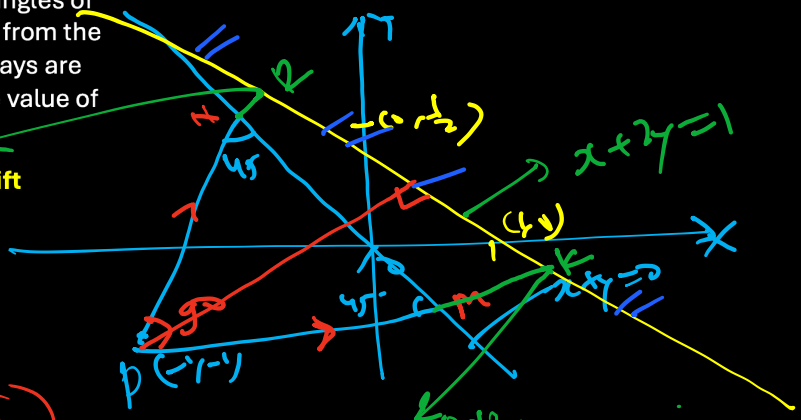
From the point $(-1,-1)$, two rays are sent making angles of $45°$ with the line $x+y=0$. These rays get reflected from the mirror $x+2y=1$. If the equations of the reflected rays are $ax+by=9$ and $cx+dy=7, a,b,c,d \in \mathbf{Z}$, then the value of $ad+bc$ is

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$m_1 m_2 = \frac{1+\tan\theta}{1-\tan\theta}$

$m_{PM} = 0$ & $m_{PN} = \infty$

eq of PM, $y=-1$

" " PN $\Rightarrow x=-1$

$K(3,-1)$ $P/2 \equiv (-1,1)$

for 1st Reflected ray $K(3,-1)$

$m=2,\ m_1=0,\ m_2=?$

$$\frac{m-m_1}{1+mm_1} = \frac{m_2-m}{1+mm_2}$$

$$\frac{2-0}{1} = \frac{m_2-2}{1+2m_2}$$

$$2+4m_2 = m_2-2$$

$$\boxed{m_2 = -\frac{4}{3}}$$

Eq of 1st Reflected ray

$$y+1 = -\frac{4}{3}(x-3)$$

$$3y+3 = -4x+12$$

$$\boxed{4x+3y=9}$$

$ax+by=9$

$\boxed{a=4,\ b=3}$

2nd Reflected ray

$m=2,\ m_1=\infty,\ m_2=?$

$$\frac{m-m_1}{1+mm_1} = \frac{m_2-m}{1+mm_2}$$

$$\frac{2-\frac{1}{m_1}}{\frac{1}{m_1}+2} = \frac{m_2-2}{1+2m_2}$$

$$\frac{-1}{2} = \frac{m_2-2}{1+2m_2} \Rightarrow -1-2m_2 = 2m_2-4$$

$$4m_2 = 3 \Rightarrow \boxed{m_2 = \frac{3}{4}}$$

$R \equiv (-1,1)$

$$y-1 = \frac{3}{4}(x+1)$$

$$4y-4 = 3x+3$$

$$3x-4y = -7$$

$-3x + 4y = 7$

$c = -3,\ d = 4$

$ad + bc = 4 \times 4 + 3 \times -3$

$= 16 - 9 = 7$ Ans